

Earlier, you learned that A/B testing is a way to compare two versions of something to find out which version performs better. For example, a data professional might use A/B testing to compare two versions of a web page or two versions of an online ad. You also learned that A/B testing utilizes statistical methods such as sampling and hypothesis testing.

In this reading, you'll learn more about the general purpose and design of an A/B test and how A/B testing uses statistical methods to analyze data.

Business context

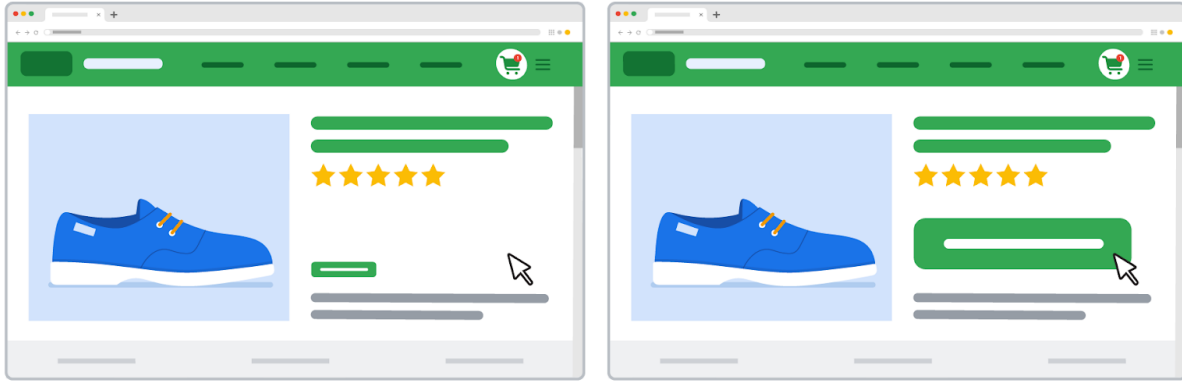
Data professionals often use A/B testing to help stakeholders choose the best design for a website or app to optimize marketing, increase revenue, or enhance customer experience. In practice, A/B testing involves randomly selecting a sample of users and dividing them into two groups (A and B). The two groups visit different versions of a company's website. The two versions are identical except for a single design feature. For instance, the "Purchase" button on Group A's version might have a different size, shape, or color than the "Purchase" button on Group B's version. An A/B test uses statistical analysis to determine whether the change in the feature (e.g., a larger button) affects user behavior for a specific metric. A data professional might use an A/B test to analyze one of the following metrics:

- *Average revenue per user*: How much revenue does a user generate for a website?
- *Average session duration*: How long does a user remain on a website?
- *Click rate*: If a user is shown an ad, does the user click on it?
- *Conversion rate*: If a user is shown an ad, will that user convert into a customer?

Let's explore an example to get a better understanding of how A/B testing works.

Example: Average revenue per user

Imagine you're a data professional who works for an online footwear retailer. The company is trying to grow its business and is researching the average revenue per user on its website. Your team leader asks you to conduct an A/B test to determine whether increasing the size of the "Purchase" button has any effect on average revenue. You randomly select a sample of users and divide them into two groups, A and B. Group A visits the standard version of the company website. Group B visits a version of the website that is identical to the standard version except for the larger "Purchase" button. You run the test online and collect your sample data. The results indicate that average revenue per user is higher for Group B. Finally, you conduct a two-sample hypothesis test to determine whether the observed difference in average revenue is statistically significant or due to chance.



A typical A/B test has at least three main features:

1. Test design
2. Sampling
3. Hypothesis testing

Let's examine each feature in more detail using our example.

Test design

First, let's discuss the fundamental design of an A/B test.

Randomized controlled experiment

An A/B test is a basic version of what's known as a randomized controlled experiment. In a **randomized controlled experiment**, test subjects are randomly assigned to a control group and a treatment group. The **treatment** is the new change being tested in the experiment. The **control group** is not exposed to the treatment. The **treatment group** is exposed to the treatment. The difference in metric values between the two groups measures the treatment's effect on the test subjects.

Note: Ideally, exposure to the treatment is the only significant difference between the two groups. This test design allows researchers to control for other factors that might influence the test results and draw causal conclusions about the effect of the treatment.

In our example, group A is the control group, group B is the treatment group, and the treatment is displaying a larger "Purchase" button. Users in the control group (A) visit the standard version of the company's website. Users in the treatment group (B) visit an alternative version with a larger "Purchase" button (i.e., are exposed to the treatment). By making the website versions for A and B identical except for the size of the "Purchase" button, you minimize the chance that any observed difference in average revenue is due to other features such as page layout or background. This allows you to measure the effect of

the larger button by comparing the difference in average revenue per user for group A and group B.

Randomization, or randomly assigning test subjects to the control group or treatment group, also helps control the potential effect of other factors on the outcome of the experiment. In practice, many different factors might influence whether a user clicks the “Purchase” button or not. For example, perhaps super wealthy users are much more likely to make purchases in general, regardless of button size. If your treatment group consists *only* of super wealthy users, you won’t get valid test results. Any observed increase in average revenue might be due to wealth, not to the larger size of the “Purchase” button (the factor you’re interested in testing). Randomization helps minimize the chance that other factors, such as wealth, will significantly influence your results on average.

Sampling

Random selection helps you create a representative sample that reflects the characteristics of the overall user population. In our example, this is the population of online customers of the company you work for. Using a representative sample for your A/B test will give you valid results that are generalizable, or applicable to the overall population.

You’ll also need to choose a sample size that is appropriate for your A/B test. The larger the sample size, the more precise the results, and the more likely you’ll get results that are statistically significant when there is a difference between group A and group B. However, working with large samples can be expensive and time-consuming. Data professionals determine sample size based on both the goal of the analysis and their available budget.

Hypothesis testing

For the purpose of our example, let’s say you run the online test, collect your data, and discover that group B has a higher average revenue per user than group A. Recall that group B is the treatment group (larger “Purchase” button), and group A is the control group. The next step is to determine whether the observed difference in your data is statistically significant or due to chance. A/B tests use two-sample hypothesis tests to draw conclusions about statistical significance. To determine whether the observed difference in average revenue per user is statistically significant, you conduct a two-sample t-test. You formulate your hypotheses as follows:

- **H₀:** There is no difference in average revenue per user between A and B
- **H_a:** There is a difference in average revenue per user between A and B

Results

Based on the results of your t-test, you reject the null hypothesis and conclude that the observed increase in average revenue per user is statistically significant.

The results of your A/B test help you decide whether or not to recommend a design change for your company's website. In this case, when you present your results to company stakeholders, you suggest implementing the larger "Purchase" button to increase average revenue per user going forward.

Key takeaways

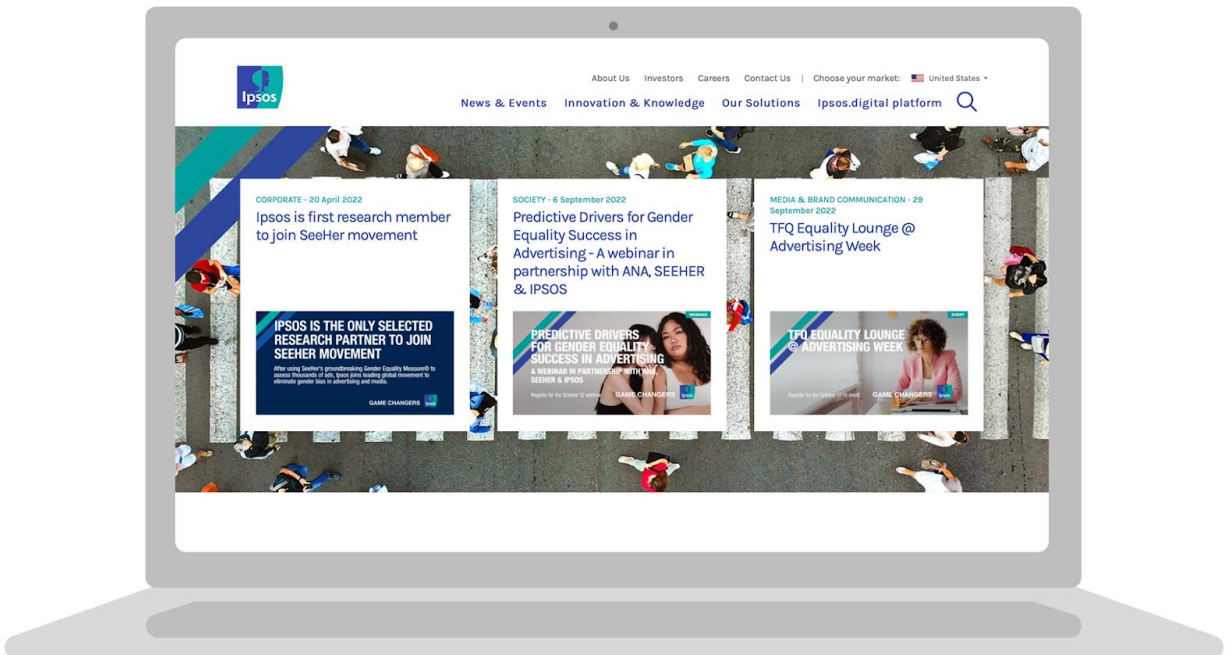
A/B testing is one of the most popular applications of statistics for business purposes. Data professionals use A/B testing to help business leaders optimize product performance, improve customer experience, and grow their online business. Understanding the general purpose and design of an A/B test will be useful in your future career as a data professional.

Previously, you learned that A/B testing is a way to compare two versions of something to find out which version performs better. For example, a data professional might use A/B testing to compare two versions of a web page or two versions of an online ad. You also learned that A/B testing utilizes statistical methods such as sampling and hypothesis testing. This case study describes how Ipsos used A/B testing to compare two different online ad formats: ads presented in a sequenced narrative vs. a traditional 30-second ad presented multiple times. You'll learn how data-driven market research reveals important insights about the impact of different ad formats on the effectiveness of a digital ad campaign.



Company background

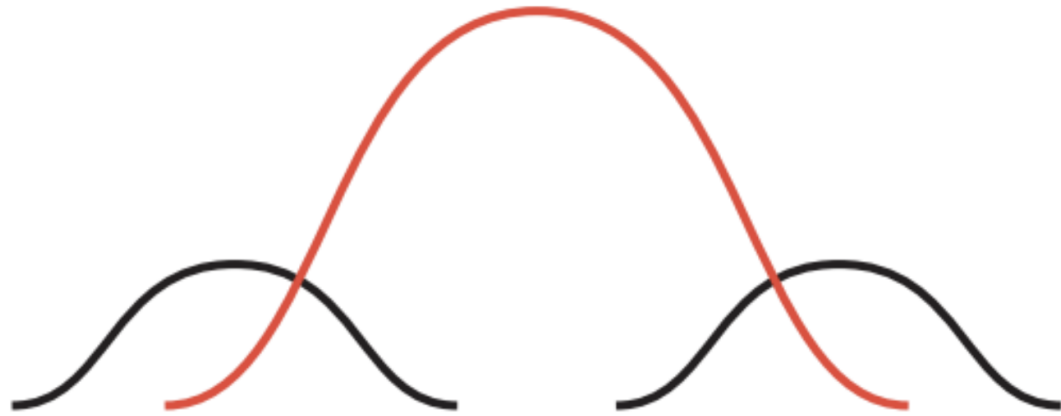
Ipsos is a full-service market research company. Founded in France in 1975, Ipsos is now a global company with 18,000 staff operating in 90 countries. Ipsos provides research services across numerous private and public sector domains. These services include brand building; advertising effectiveness; product development; reputation; customer and user experience; and public opinion, election, and crisis management. Ipsos uses a combination of data sources for their research, from primary data collection to social listening, mobility, and satellite imagery.



Project background

Ipsos' client for this project was an online media company that allows users to post their own video content. The media company wanted to help its own clients - the advertisers on their platform - create the most effective ads. In particular, they wanted to find out whether investing in sequenced ads would increase the likelihood of viewers recalling an ad and purchasing a product. Video ad sequencing lets advertisers show ads in an order based on the most engaging and memorable story structures. The media company commissioned Ipsos to conduct research to measure the impact of five different sequence structures on brand lift.

Note: For the purpose of this case study, we will focus on only one sequence structure: Tease, Amplify, Echo. This sequence starts with a short ad to spark viewer curiosity (Tease); then, it moves on to a longer ad with more information to secure viewer engagement (Amplify); finally, it ends with a shorter ad that recaps the story and spurs viewers to action (Echo).



Tease, Amplify, Echo

Graph with three overlapping curves represents the structure of the Tease, Amplify, Echo ad sequence: first, short video; second, long video; third, short video.

Project framework

Ipsos developed the following research question to guide their project: Does an ad sequence with the Tease, Amplify, Echo structure increase ad recall and purchase intent compared to repeated viewings of a traditional 30-second ad?

Ipsos' initial hypothesis was that ads presented in a sequenced narrative would be more effective than repeated viewings of a traditional ad. To test this hypothesis for these two advertising approaches, Ipsos conducted an A/B test. The A/B test set up an experiment for two groups of users: one group was shown the Tease, Amplify, Echo ads and the other group was shown a traditional ad multiple times. In each case, the different ad formats were based on the same brand content. You'll learn more about the details of the testing process in what follows.

The challenges

At the outset of the project, Ipsos identified two main challenges. The first challenge involved properly designing the A/B test. The second challenge involved creating the test ads in the appropriate test environment.

Test design

Ipsos's primary concern was for the results of the A/B test to be generalizable, or applicable to the overall population of the media company's users. In other words, Ipsos wanted to make valid inferences about the larger user population based on the smaller sample of test participants. To obtain valid test results, Ipsos needed to do the following;

1. Create a representative sample of test participants that mirrored the overall population of the media company's users.
2. Create an online test environment that mirrored the media company's online environment. This also implied creating test ads from multiple brands to reproduce the diversity of ads featured on the media company's platform.

The approach

Despite these challenges, Ipsos conducted the A/B test and achieved their research goals. Ipsos' successful approach to their project included the following elements:

- Team
- Sampling
- Testing process
- Hypothesis testing

Team

To build an effective team, Ipsos created a cross-functional operation, including video production to create test ads based on the Tease, Amplify, Echo structure, and technology to build a realistic online test environment.

To facilitate collaboration among project participants, Ipsos outlined a clear set of editing rules and organized a shared site to house links to videos and edit notes. This allowed rapid feedback and adjustment throughout the development process. Finally, Ipsos had senior client service project managers own and monitor the project design and workflow from beginning to end.

Sampling

Ipsos used random selection from consumer panels to generate a representative sample that accurately reflected the characteristics of the overall user population. Ipsos also made sure that each test group included the same proportion of respondents for key categories such as age and gender. Further, Ipsos used a relatively large sample size of 7,500 respondents to obtain more precise results.

Testing process

To get valid test results, Ipsos conducted A/B testing in an online environment where respondents used the media company's platform as they typically would in their everyday lives. To mirror the diversity of ads on the platform, Ipsos developed test ads across 30 brand categories, from airline tickets to fast food to laundry detergent.

The testing process was organized in the following way:

Surveys were administered online via respondents' smartphones in November and December of 2018. After initial screening, respondents were taken to a browser-based version of the platform where they were free to search for and watch videos as they normally would. Ipsos dynamically inserted test ads at the beginning of the videos selected by respondents in the live test environment. After the browsing session, respondents completed a survey to measure brand lift for ad recall and product intent.

Hypothesis testing

The survey data indicated that the Tease, Amplify, Echo ad sequence led to higher levels of ad recall and purchase intent among respondents than repeated viewings of a traditional ad. To determine whether their observed results were statistically significant, Ipsos conducted a two-sample t-test for each category: one for ad recall and one for purchase intent. They formulated their hypotheses as follows:

- **H₀:** There is no difference in ad recall between sequenced ads and a repeated traditional ad.
- **H_a:** There is a difference in ad recall intent between sequenced ads and a repeated traditional ad.
- **H₀:** There is no difference in purchase intent between sequenced ads and a repeated traditional ad.
- **H_a:** There is a difference in purchase intent between sequenced ads and a repeated traditional ad.

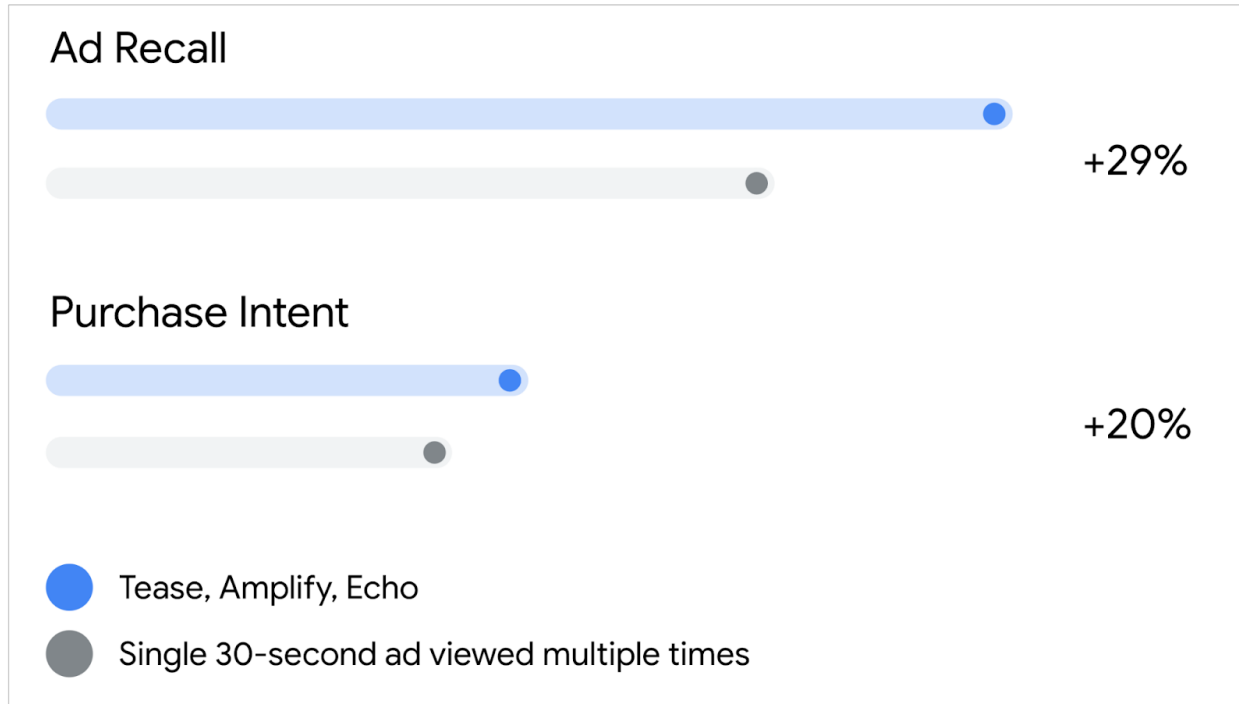
For both tests, Ipsos rejected the null hypothesis. They concluded that there are statistically significant and substantively meaningful differences in ad recall and purchase intent between sequenced ads and a repeated traditional ad.

The results

The results of the A/B test indicated that ad sequencing works!

The Tease, Amplify, Echo ad sequence had a significantly greater impact on ad recall and purchase intent than repeated viewings of a traditional ad. For example, across all product

categories, 54% of viewers recalled the ad after being exposed to the “Tease, Amplify, Echo” sequence as compared to 42% with a repeated traditional ad. Further, 30% of viewers expressed intent to purchase after being exposed to the “Tease, Amplify, Echo” ad sequence as compared to 25% with a repeated traditional ad.



Bar chart showing the increase in ad recall and purchase intent for the Tease, Amplify, Echo ad sequence as compared with a repeated 30-second ad.

Overall, the results suggest that advertisers should invest in ad sequencing for their digital campaigns to increase brand lift.

Conclusion

This case study on Ipsos’ A/B testing demonstrates the power of data-driven research to generate key business insights. The results of the A/B test clearly show how ad sequencing increases ad recall and purchase intent compared to repeated viewings of a traditional ad. Ipsos’s research on the benefits of ad sequencing helped the media company improve the experience and performance of the advertisers on their platform and added value to the media company’s brand.